A Major Project Report

(Phase-I)

On

AI-Based Crop Disease Detection and Smart Farming System

Submitted in partial fulfillment of the

requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the Phase-I report of the project entitled "AI-Based Crop Disease Detection and Smart Farming System", in partial fulfillment of the requirements for the degree of Bachelor of Technology in the School of Computing Science and Engineering, IILM University, Greater Noida, is an original work carried out by us under the supervision of **Dr. Harendra Singh**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Indian agriculture suffers significant yield losses every year due to late or incorrect identification of crop diseases, coupled with inefficient irrigation and fertilizer practices. This project proposes an AI-Based Crop Disease Detection and Smart Farming System that uses deep learning, specifically Convolutional Neural Networks (CNN), to identify plant diseases from leaf images captured through a smartphone or field camera. The system further integrates IoT sensor data (soil moisture, temperature, and humidity) to provide real-time recommendations for irrigation, fertilization, and pest control through a simple mobile/web interface. A transfer-learning model (based on architectures such as MobileNetV2/ResNet) is trained on the PlantVillage dataset to classify healthy and diseased leaves across multiple crop species with high accuracy. The proposed system aims to reduce dependency on manual expert inspection, minimize crop loss, optimize resource usage, and empower small and marginal farmers with affordable, accessible, and timely agricultural intelligence. This Phase-I report covers the problem definition, literature survey, objectives, scope, and feasibility study of the proposed system.

1. Problem Definition

Agriculture forms the backbone of the Indian economy, employing nearly half of the country's workforce. However, crop diseases destroy an estimated 20-40% of global agricultural productivity every year, according to FAO reports. Farmers, especially in rural and remote areas, largely rely on visual inspection and traditional knowledge to identify plant diseases, which is often slow, subjective, and error-prone. By the time visible symptoms are recognized and diagnosed correctly, the disease may have already spread across the field, causing irreversible yield and economic loss.

Access to agricultural experts and pathologists is limited, particularly for small and marginal farmers, and laboratory-based diagnosis is expensive and time-consuming. In addition, farmers frequently over-use pesticides and fertilizers due to lack of precise, data-driven guidance, which increases input costs, degrades soil health, and harms the environment. There is, therefore, a strong need for an intelligent, low-cost, and easy-to-use system that can automatically detect crop diseases at an early stage and simultaneously guide farmers on optimal irrigation and fertilization practices.

This project addresses this gap by proposing an AI-based system that combines image-based deep learning for disease detection with IoT-based environmental sensing for smart farming recommendations, thereby making precision agriculture accessible to farmers with minimal technical expertise.

2. Literature Survey

An extensive review of existing research was carried out to understand current approaches in AI-based plant disease detection, deep learning image classification, and IoT-enabled smart farming.

Mohanty et al. [1] used a deep convolutional neural network trained on the PlantVillage dataset of 54,306 leaf images spanning 14 crop species and 26 diseases, achieving 99.35% accuracy on held-out test data. The study demonstrated the feasibility of smartphone-assisted disease diagnosis but noted a significant drop in accuracy when the model was tested on field images with complex backgrounds, unlike the plain-background lab images used for training.

Ferentinos [2] developed and compared several CNN architectures (AlexNet, VGG, GoogLeNet) for plant disease detection using an open database of 87,848 leaf images covering 25 plant species, obtaining a best accuracy of 99.53%. The work highlighted that model performance degraded considerably for images captured under real cultivation conditions rather than controlled laboratory settings, motivating the need for field-representative datasets.

Sladojevic et al. [3] proposed a CaffeNet-based deep CNN capable of recognising 13 different plant diseases from healthy leaves, reporting an average precision of 96.3%. The authors emphasised that the model's reliability depended heavily on the diversity and quality of the training images and suggested that data augmentation could improve generalisation to unseen conditions.

Too et al. [4] performed a comparative evaluation of fine-tuned deep architectures, including VGG16, Inception V4, ResNet, and DenseNet, on the PlantVillage dataset. DenseNet achieved the highest testing accuracy of 99.75% while requiring fewer trainable parameters. The study concluded that deeper, densely connected networks generalise better but require careful tuning of computational resources for deployment on low-power devices.

Kamilaris and Prenafeta-Boldú [5] conducted a systematic survey of 40 deep learning research efforts applied to agriculture, covering crop classification, weed and disease identification, and yield prediction. The review found that deep learning models consistently outperformed traditional image-processing techniques but identified the scarcity of large, field-collected, labelled agricultural datasets as a major bottleneck for real-world adoption.

Hughes and Salathé [6] introduced the PlantVillage repository, an open-access collection of over 50,000 expert-annotated leaf images, created to support the development of mobile disease-diagnostic tools for farmers in regions with limited access to agricultural extension services. This dataset has since become the benchmark for most subsequent plant disease classification research, including several works reviewed in this survey.

Brahimi et al. [7] applied deep CNNs specifically to tomato leaf disease classification using 14,828 images spanning nine disease classes, achieving over 99% accuracy, and additionally used saliency-map visualisation to highlight the diseased regions influencing the model's decision. The work showed that visual explanation techniques can increase farmer trust in automated diagnosis but noted the dataset was still lab-captured rather than field-acquired.

Fuentes et al. [8] proposed a real-time deep-learning detector combining Faster R-CNN, SSD, and R-FCN meta-architectures to simultaneously localise and classify nine tomato diseases and pests directly from field images, reporting a mean average precision of 85.98%. The study is notable for using naturally captured field images rather than clean lab photographs, but flagged class imbalance and illumination variation as recurring sources of false detections.

Ramcharan et al. [9] applied transfer learning with Inception v3 to detect cassava diseases and pests directly from mobile-phone images captured in Tanzanian farm fields, achieving accuracies between 88% and 98% across classes. The study demonstrated that a lightweight, offline-deployable model could run on low-cost smartphones, making it relevant for smallholder farmers in low-connectivity rural regions.

Barbedo [10] argued that classifying whole leaf images conflates multiple disease symptoms and background noise, and instead proposed training CNNs on individual lesions and spots segmented from the leaf surface. This lesion-level approach improved recognition of diseases with subtle or overlapping symptoms but required an additional, computationally expensive segmentation step prior to classification.

Picon et al. [11] trained deep CNNs on a large field-acquired dataset of wheat images captured using mobile devices under uncontrolled lighting and background conditions, incorporating contextual metadata such as crop growth stage. The model achieved balanced accuracy above 96% and outperformed models trained solely on lab images, reinforcing the importance of field-representative training data for real deployment.

DeChant et al. [12] combined a three-stage deep learning pipeline with UAV-captured field imagery to automatically identify maize plants infected with northern leaf blight, reporting 96.7% accuracy. The study illustrated how deep learning can be extended beyond handheld leaf photography to aerial, large-scale field monitoring, though it required substantial computational pipeline complexity for the multi-stage detection process.

Ayaz et al. [13] surveyed IoT-based smart agriculture systems, reviewing sensor networks, communication protocols, and data-analytics platforms used for soil, weather, and crop-health monitoring. The survey concluded that IoT integration enables real-time, data-driven farm management decisions but identified interoperability between heterogeneous sensor platforms and limited rural network connectivity as ongoing implementation challenges.

Elijah et al. [14] reviewed the application of IoT and big-data analytics in agriculture, covering precision irrigation, livestock monitoring, and supply-chain traceability. The paper found that combining sensor data with predictive analytics improves resource-use efficiency and yield forecasting, while also cautioning that high deployment costs and lack of farmer digital literacy restrict adoption in developing regions.

FAO [15] reports that plant pests and diseases are responsible for the loss of up to 40% of global food crop production annually, with significant economic impact on smallholder farmers who lack timely access to diagnostic expertise. This underscores the practical motivation for accessible, automated, AI-based disease-detection tools of the kind proposed in this project.

Research Gap

- Most existing CNN models are trained and validated on lab-captured, plain-background leaf images (e.g., PlantVillage), and their accuracy drops significantly on real, cluttered field images.
- Few systems integrate disease detection with real-time environmental (IoT) data to provide holistic farming recommendations rather than disease labels alone.
- Existing solutions are often not optimized for low-resource, offline, or low-bandwidth rural environments, limiting accessibility for small farmers.
- There is limited work on providing actionable, localized advisory (dosage, treatment, irrigation schedule) rather than just a diagnostic label.

3. Objectives

The primary aim of this project is to design and develop an intelligent, accessible system for early crop disease detection and smart farming guidance. The specific objectives are:

1. To develop a deep learning (CNN-based) model capable of accurately classifying healthy and diseased crop leaves from images across multiple crop species.
2. To design a mobile/web-based interface that allows farmers to capture or upload leaf images and receive instant diagnosis along with recommended remedies.
3. To integrate IoT sensors (soil moisture, temperature, humidity) for real-time environmental monitoring of the farm.
4. To build a recommendation engine that suggests optimal irrigation schedules, fertilizer usage, and pesticide dosage based on detected disease and sensor data.
5. To evaluate the model's performance using standard metrics (accuracy, precision, recall, F1-score) and optimize it for deployment on low-resource devices.
6. To provide multilingual, farmer-friendly outputs to ensure usability among non-technical rural users.

4. Scope

The scope of this project encompasses the design, development, and preliminary evaluation of an AI-based crop disease detection and smart farming advisory system. It includes the following aspects:

- Disease detection support for major crops such as tomato, potato, maize, and wheat using publicly available and field-collected image datasets.
- Development of a CNN/transfer-learning based image classification model deployable via a web or Android application.
- Integration of basic IoT sensor modules for environmental parameter monitoring on a sample smart farming test-bed.
- A recommendation module offering general treatment and cultivation guidance based on the detected disease.
- The system is intended primarily as an academic prototype/proof-of-concept and is not a certified replacement for professional agronomist or laboratory diagnosis.
- Large-scale commercial deployment, hardware manufacturing, and support for all crop varieties/diseases are considered beyond the current scope and are identified as future work.

5. Feasibility Study

5.1 Technical Feasibility

The project relies on well-established and freely available technologies: Python, TensorFlow/Keras or PyTorch for model development, OpenCV for image pre-processing, and Flask/Django or a Flutter/Android front-end for the user interface. Public datasets such as PlantVillage provide sufficient labeled data to train an initial model. Low-cost IoT hardware (ESP32/Arduino with DHT11 and soil-moisture sensors) is readily available for environmental sensing. Given the availability of these tools, cloud GPU resources (Google Colab/Kaggle), and existing literature/pretrained models, the project is technically feasible within an academic setting.

5.2 Economic Feasibility

The project uses open-source software and libraries, eliminating licensing costs. Hardware for the IoT prototype (microcontroller and sensors) is low-cost, typically under a few thousand rupees, making it affordable for an academic project. Cloud-based training resources are available free of cost or at minimal cost for student use, keeping the overall budget well within institutional project funding limits.

5.3 Operational Feasibility

The proposed mobile/web interface is designed to be simple and intuitive, requiring only that a user capture or upload a leaf photo. This makes it operable by farmers with minimal digital literacy. The system can function with intermittent internet connectivity by supporting on-device (edge) inference for the classification model, and the IoT module operates autonomously, requiring minimal manual intervention once set up.

5.4 Schedule Feasibility

The project is planned across two phases: Phase-I (current phase) covers problem identification, literature survey, requirement analysis, and feasibility study. Phase-II will involve dataset preparation, model development and training, IoT integration, application development, testing, and final evaluation, to be completed within the remaining project timeline as per the academic calendar.

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